QUESTION 5: SOC PLAYBOOK / RUNBOOK

DG Interns Hub Cybersecurity Internship – Week 5 Submission

Student Name: Gunjan **Track:** SOC Analyst Tier 1 **Topic:** Standard Operating Procedures & Response Automation

1. WHAT IS A SOC PLAYBOOK / RUNBOOK?

A **SOC Playbook** (also known as a Runbook or Standard Operating Procedure - SOP) is a documented, step-by-step workflow that guides analysts on how to investigate, verify, contain, and remediate specific security scenarios (such as Phishing, Ransomware, Malware Execution, or Unauthorized Access).

2. WHY SOC TEAMS RELY ON PLAYBOOKS

Playbooks serve as operational blueprints for security teams. Modern SOCs handle complex cyber threats daily; playbooks ensure the response is systematic rather than reliant on individual memory or guesswork.

> **Standard Procedures & Consistency:** Ensures every analyst—regardless of shift time or experience—follows the exact same investigation baseline for identical alerts. **Faster Incident Response (Lower MTTR):** Eliminates hesitation during active security incidents, significantly reducing Mean Time to Respond (MTTR). **Reduced Human Error:** Prevents analysts from forgetting critical diagnostic steps (e.g., failing to check email headers, check IP reputation, or log ticket details). **Onboarding & Training Tool:** Enables junior (L1) analysts to quickly learn operational workflows and handle real-world alerts effectively. **SOAR Automation Ready:** Serves as the logic template for Security Orchestration, Automation, and Response (SOAR) playbooks.

3. PRACTICAL EXAMPLE: PHISHING RESPONSE PLAYBOOK WORKFLOW

Below is a standard step-by-step SOC playbook executed when an employee reports a suspicious email:

1. **Initial Triage & Header Inspection:** Inspect raw `.eml`/`.msg` headers. Verify SPF, DKIM, and DMARC alignment status to check for spoofing.
2. **Indicator Extraction:** Extract sender domain, embedded URL links, IP addresses, and attached file hashes.
3. **Reputation Analysis:** Check URLs and file hashes on Threat Intelligence platforms (VirusTotal, AbuseIPDB, Any.Run sandbox).
4. **Scope Impact Assessment:** Search email gateway logs to see if other internal employees received the same email or clicked the malicious link.
5. **Containment & Remediation:** Purge email from all employee inboxes, block domain/IP at firewall, and force password reset for affected users.

4. PRACTICAL SOC PLAYBOOK SCENARIO

Scenario: Suspicious PowerShell Execution Playbook

Alert: EDR fires for `powershell.exe -enc ...` on a corporate workstation.

Playbook Execution:

Step 1: Decode Base64 string using CyberChef. **Step 2:** Check if script attempts external download or process injection. **Step 3:** If malicious, execute host isolation command in EDR immediately. **Step 4:** Escalate to L2 Incident Response with decoded payload and process tree screenshot.

5. TERMINOLOGY & REFERENCES

SOAR: Security Orchestration, Automation, and Response—software that automates playbook steps.
MTTR: Mean Time to Respond—metric tracking how quickly incidents are resolved.

References:

• CISA Incident Response Playbooks: **https://www.cisa.gov/**
• NIST SP 800-61 Rev. 2 Guide to Incident Handling: **https://csrc.nist.gov/**